

Quantum Security

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Presentation Agenda

- Intro to Quantum Encryption and how it relates to Network Security
 - Classical computing systems
 - Classical Encryption
 - Quantum computing systems
 - Quantum Threat
 - Quantum Encryption
- Quantum Encryption for Network Security
 - Post Quantum Cryptography
 - Quantum Key Distribution

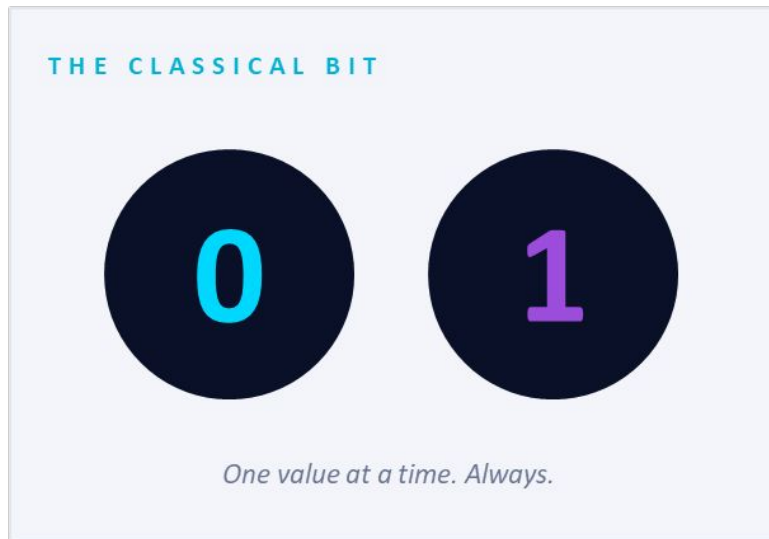
The Future Work

- Quantum Encryption as a use case scenario for an experimental Quantum Network
 - How do the concepts of post-quantum cryptography and Quantum Key Distribution be integrated into the emerging Quantum Network of entangled qubits?
- Justification Slides

Classical Computing

Classical computing systems

Bits are deterministic — they hold one value at a time. Encryption relies on problems classical machines find hard.



■ Deterministic logic

Every operation has one definite outcome — perfect for predictable computation.

■ Sequential by nature

Problems are solved step-by-step; speed scales with clock rate and parallel cores.

■ Security through math

Encryption rests on problems too costly for classical machines — factoring huge integers, discrete logs.

Classical Encryption

Modern encryption is built on problems classical computers cannot solve in any reasonable time.

ASYMMETRIC

RSA

Typical key: 2048-bit

Built on the difficulty of factoring the product of two large primes. The workhorse of TLS key exchange for decades.

SYMMETRIC

AES

Typical key: 256-bit

Used to encrypt data in transit and at rest. Relies on the brute-force infeasibility of guessing a shared secret key.

ASYMMETRIC

ECC

Typical key: 256-bit

Elliptic-curve cryptography — smaller keys, same strength as RSA. Powers most modern HTTPS handshakes.

PRIME · P (1024 BITS)

```
C7A3F1D9 58B2E40C 4F1908B2 605A77E1 9C4083
A5 7E11F2D8 B36C5491 E028AF73 4D9E1C82 8B5
7F386 1A4D9B72 C5E809F1 6E7B5002 9F138C44
70A2E519 D1B6C0F BC95A1E7 30482D06 51F9C7
0E 9A6D2830 C48E12F5 8A435D7C 6B2918EF 3D5
09A41 9F2E8074 5D1AB623 8E04F619 2C97D31A
4B58E12D 7039C2F6 81DA5847 905EA13C
```

a randomly-generated 1024-bit prime

×

PRIME · Q (1024 BITS)

```
D5419E8B 7C20A634 81F59307 60E2A4C9 3B5847
F1 9E62D88A 4C7B1F30 52809E61 AF03C481 7E2
9B5D2 8F6B41C7 35E9A8B2 81D40627 5C9EF813
4A7B920C 6D3F501E E5942018 7A3CD41B 903F71
E8 3C48B0A3 1F69C408 83D2E517 6A901C24 7F5
E8380 4D29C6E5 B3801F47 A6052E09 1C82B70F
85629301 D74EAB12 4F09C5E6 38B71042
```

a second random 1024-bit prime

=

MODULUS · N = P · Q (2048 BITS)

```
A7F08B14 EC9521D6 3F417A95 80E28D3C 7C61F4
98 5B002A36 9E84C175 30F6B14E 82A91D07 4FB
C358E 6092A1D7 5C803F46 BF3D5908 12E7A4C6
5D81F309 7A26C408 3E5B0918 D672CA5A 8F19E0
34 60B7A905 21C8E407 9358F1A4 6BD20E5F C70
48391 4A91C603 E58F1027 706BD24E 9F134825
C50A7E96 1D804B53 8FA729C0 4E17B628 9A2C56
D0 7E48B315 02FCE1A9 6D5403B7 E91CA07F D02
B5083 91A4E72F 8B5061C2 4D3ABE15 70CF9B26
81E0345D 7C26B498 F1D804A3 25E6798B 4C863F
1A 90E85206 5B0A3BF2 7C4F1895 30F6271B D1A
09017 8F6C83D1 5E9B002A 84C71F36 D200A85E
3C70B541 E809C26F 5D017A30 4B921F8E B602C5
3F 8A047D91 6E382B14 95C720A0
```

factoring this 617-digit number is computationally infeasible classically

Classical Encryption

Modern encryption is built on problems classical computers cannot solve in any reasonable time.

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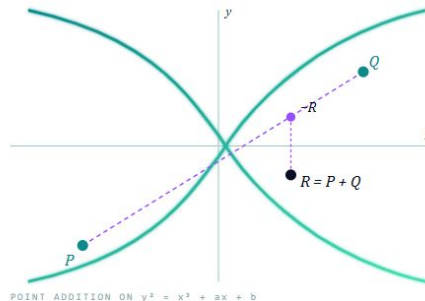
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Quantum Threat

Two algorithms reshape what's possible — and what's safe.

Shor's Algorithm

1994

THREATENS

RSA · ECC · Diffie-Hellman

BROKEN

Factors large integers and solves discrete logarithms in polynomial time. The mathematical foundation of asymmetric encryption collapses on a sufficiently powerful quantum computer.

Grover's Algorithm

1996

THREATENS

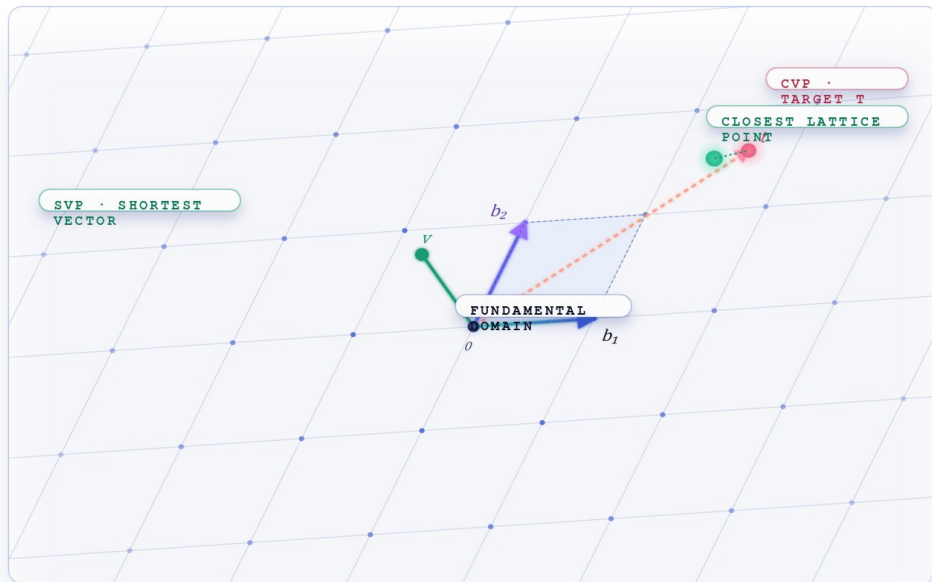
AES · SHA-256 · symmetric ciphers

WEAKENED

Provides a quadratic speed-up on brute-force search. AES-128 effectively offers only 64 bits of security — doubling key sizes (AES-256) restores safety.

HARVEST NOW, DECRYPT LATER · Encrypted traffic captured today can be broken once quantum hardware matures.

Post Quantum Cryptography



LATTICE · DEFINITION

$$\Lambda = \{ \sum_i z_i b_i : z_i \in \mathbb{Z} \}$$

An infinite, regular grid of points generated by a basis $B = (b_1, \dots, b_n)$. The same lattice can be described by many bases — but only some make hard problems tractable.

LWE · LEARNING WITH ERRORS

$$b = A \cdot s + e \pmod{q}$$

Recovering the secret s from noisy linear samples (A, b) is as hard as worst-case lattice problems — even for a quantum adversary.

WHY IT RESISTS QUANTUM ATTACKS

SVP

Shortest non-zero vector in Λ

CVP

Closest lattice point to a target t

BDD

Bounded-distance decoding

Kyber

Dilithium

Falcon

NTRU

FrodoKEM

The **Lattice** Behind Quantum-Safe Cryptography

Post Quantum Cryptography

A public key $\mathcal{P}(x_1, \dots, x_n)$ is m quadratic polynomials

Encryption / signing is a one-way evaluation of a public quadratic system over a finite field $\mathbb{F}_q$. Inverting it — solving a generic system of multivariate quadratic equations — is NP-hard, and stays hard for quantum computers.

$$y_1 = x_1x_3 + x_2^2 - x_2x_5 + x_4 + 3 \pmod{q}$$

$$y_2 = x_1x_2 + x_3x_4 - x_5^2 + 2x_1 - 5$$

$$y_3 = x_2x_4 + x_3^2 - x_1x_5 + 7x_3 - x_4$$

$\vdots$

$$y_m = x_1x_4 - x_2x_3 + x_5^2 + 4x_2 + 1$$

SECRET · S

$$x = S(x')$$

Affine map on inputs

CENTRAL · F

$$y' = F(x')$$

Easy-to-invert quadratic map

SECRET · T

$$y = T(y')$$

Affine map on outputs

MQ · THE HARD PROBLEM

$$\text{Find } x \in \mathbb{F}_q^n : \mathcal{P}(x) = y$$

Solving a random system of m quadratic equations in n variables over a finite field is **NP-hard**. No known quantum algorithm gives a sub-exponential speed-up.

TRAPDOOR

$$\mathcal{P} = T \circ F \circ S$$

The designer hides an easily-invertible F inside two secret affine layers S and T . Knowing (S, F, T) means signing or decrypting is fast; without them the system looks random.

WHY IT RESISTS QUANTUM ATTACKS

MQ

Multivariate Quadratic system

IP

Isomorphism of polynomials

MINIRANK

Find low-rank matrix combination

Rainbow

UOV

MAY

GeMS

HFE

The Multivariate Equations Behind Quantum-Safe Signatures

Post Quantum Cryptography

A three-step protocol: **KeyGen** → **Encaps** → **Decaps**

Two parties agree on a shared secret K by exchanging only public values.

Security reduces to the **Module Learning With Errors (M-LWE)** problem on structured lattices — believed hard even for a quantum adversary.

ALICE · RECEIVER

① KEYGEN

$$A \leftarrow R_{q \times k}^{k \times k}, s, e \leftarrow \chi$$

$$t = A \cdot s + e$$

Publish public key $\text{pk} = (A, t)$; keep secret $\text{sk} = s$.

③ DECAPS

$$m' = \text{Decode}(v - s^T \cdot u)$$

Recover the message and derive the shared key $K = H(m')$.

PK

CT = $\begin{pmatrix} u \\ v \end{pmatrix}$

SHARED
SECRET
 $K = H(m)$

BOB · SENDER

② ENCAPS

$$m \leftarrow \mathcal{M} \subseteq \{0,1\}^{256}$$

$$r, e_1, e_2 \leftarrow \chi$$

$$u = A^T \cdot r + e_1$$

$$v = t^T \cdot r + e_2 + \lfloor q/2 \rfloor \cdot m$$

Send ciphertext $\text{ct} = (u, v)$; derive $K = H(m)$.

M-LWE · THE HARD PROBLEM

$$(A, A \cdot s + e) \approx_c (A, u)$$

Distinguishing noisy module-lattice samples from uniform is conjectured hard — even for quantum computers. Module structure brings compact keys and fast arithmetic via NTT.

KEM · IND-CCA SECURITY

KeyGen / Encaps / Decaps

A Fujisaki–Okamoto transform lifts the IND-CPA PKE to an IND-CCA2 KEM. Implicit rejection of malformed ciphertexts protects against active and side-channel attacks.

WHY IT RESISTS QUANTUM ATTACKS

M-LWE

Module Learning With Errors

SVP

Shortest vector on lattices

NTT

Fast polynomial arithmetic

ML-KEM-
512

ML-KEM-
768

ML-KEM-
1024

FIPS
203

CRYSTALS-
Kyber

The **ML-KEM** Key Encapsulation from Module-Lattices

Quantum Computing Systems

Qubits exploit superposition and entanglement — solving certain problems exponentially faster than classical machines.



■ Superposition

A qubit can be both 0 and 1 at once. n qubits can explore 2^n states in parallel.

■ Entanglement

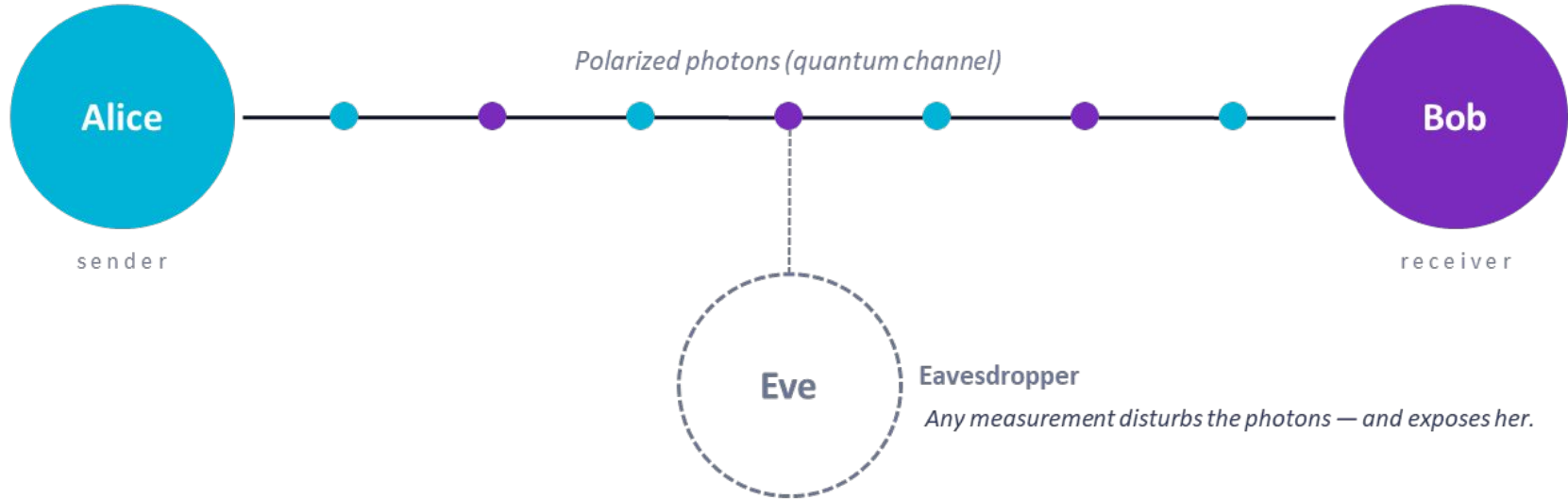
Linked qubits share state instantly — measuring one tells you about the other, however far apart.

■ Quantum algorithms

Specific problems (factoring, search) collapse from intractable to feasible — and that's the threat to encryption.

Quantum Encryption

Quantum Key Distribution uses the laws of physics — not mathematical hardness — to share a secret key.



BB84 (Bennett & Brassard, 1984) · Security from the no-cloning theorem, not from computational hardness.

Quantum Encryption (cont'd)

BB84 — Step 1: Alice Encodes & Sends

THE PROTOCOL BEGINS

Alice flips a coin n times to choose her bits, and n more times to choose the basis (+ or $\times$) for each. She then sends each bit using the chosen basis.

Bit #	1	2	3	4	5	6	7	8	9	10	11	12
Alice's bit	0	1	1	0	1	1	1	0	1	0	1	0
Alice's basis	+	+	$\times$	+	+	+	$\times$	+	$\times$	$\times$	$\times$	+
Alice sends	$\rightarrow$	$\uparrow$	$\nearrow$	$\rightarrow$	$\uparrow$	$\uparrow$	$\searrow$	$\rightarrow$	$\nearrow$	$\searrow$	$\nearrow$	$\rightarrow$

Encoding rules

Bit 0 in basis + $\rightarrow |\rightarrow\rangle$ Bit 1 in basis + $\rightarrow |\uparrow\rangle$
Bit 0 in basis $\times$ $\rightarrow |\nearrow\rangle$ Bit 1 in basis $\times$ $\rightarrow |\searrow\rangle$

Quantum Encryption (cont'd)

BB84 — Step 2: Bob Measures Randomly

ROUGHLY HALF THE BASES WILL MATCH

Bob does not know which basis Alice used, so he flips a coin n times to choose his measurement basis for each qubit, and records the outcomes.

Bit #	1	2	3	4	5	6	7	8	9	10	11	12
Bob's basis	×	+	×	×	+	×	×	+	×	×	×	+
Bob observes	↗	↑	↘	↗	↑	↗	↘	→	↘	↗	↘	→
Bob's bit	0	1	1	0	1	0	1	0	1	0	1	0

Outcome statistics

≈ 50% of the time — Bob's basis matches Alice's. His bit is *exactly Alice's*.

≈ 50% of the time — bases differ. Bob's bit agrees with Alice's only ≈ 50% of the time.

Quantum Encryption (cont'd)

Eve Tries to Intercept

WHY SHE INEVITABLY LEAVES FINGERPRINTS

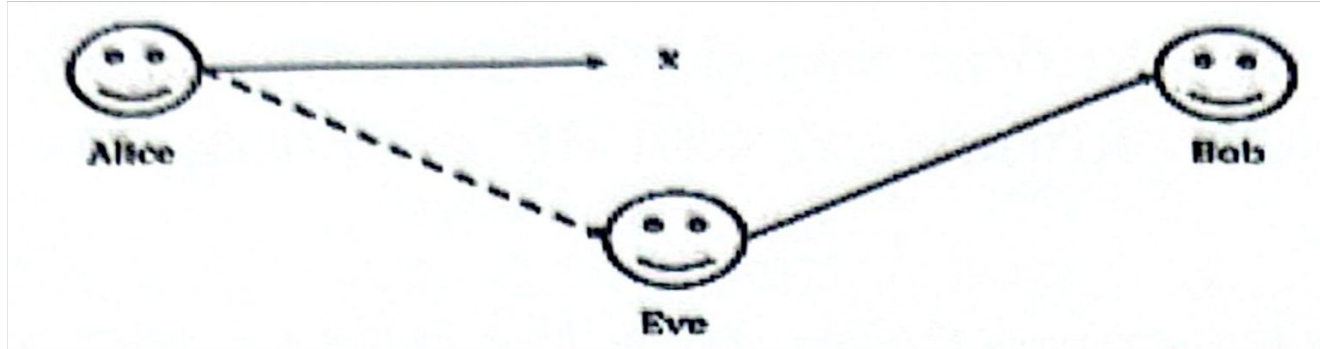


Figure 9.5 — Eve cuts the quantum line and resends her own messages

What Eve faces

- Eve also doesn't know Alice's basis — she must guess by flipping a coin.
- When she guesses wrong, she collapses the qubit into her own basis.
- **No-cloning theorem:** she cannot resend the original — only the collapsed copy.
- Bob now sees a qubit in the wrong basis half the time → his agreement with Alice drops below 50% on those bits.

Quantum Key Distribution

Two parties build a shared key from single photons

Alice sends qubits encoded in randomly chosen bases (rectilinear $|+\rangle$ or diagonal $|x\rangle$). Bob measures in randomly chosen bases. After transmission they publicly compare bases — keeping only the bits where both bases matched. Any eavesdropping disturbs the qubits and shows up as a higher QBER.



Step	1	2	3	4	5	6	7	8
Alice basis	+	x	+	x	+	+	x	x
Alice bit	0	1	1	0	1	0	1	1
Bob basis	+	+	+	x	x	+	x	+
Sifted key	0	-	1	0	-	0	1	-

BB84 - PROTOCOL

$$|\psi\rangle \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$$

Alice encodes each bit at random in one of two non-orthogonal bases. Bob measures at random. After sifting, only bases that *matched* contribute to the raw key.

EAVESDROPPING - DETECTABLE NOISE

$$QBER \gtrsim 11\% \Rightarrow \text{abort}$$

The no-cloning theorem forbids copying an unknown qubit, so any intercept-resend by Eve disturbs the statistics. The error rate on a small public sample bounds her information.

FOUNDATIONS OF QUANTUM SECURITY

NO-CLONING

Unknown qubits cannot be perfectly copied

MEASUREMENT

Disturbs incompatible bases

ENTANGLEMENT

Used by E91 / device-independent QKD

BB84

E91

SARGO

MDI-QKD

Twin-Field

BB84 Protocol

Quantum Encryption for Network Security

What this means for network security

Two parallel paths forward — one mathematical, one physical.

SOFTWARE PATH

Post-Quantum Cryptography

New math, runs on classical hardware

- Lattice-based, hash-based, and code-based algorithms
- NIST has standardized CRYSTALS-Kyber and Dilithium
- Drop-in upgrade for TLS, VPNs, and digital signatures

HARDWARE PATH

Quantum Key Distribution

Physics-based, needs special hardware

- Provably secure against any computational attack
- Requires fiber or satellite quantum channels
- Used today in finance and government networks

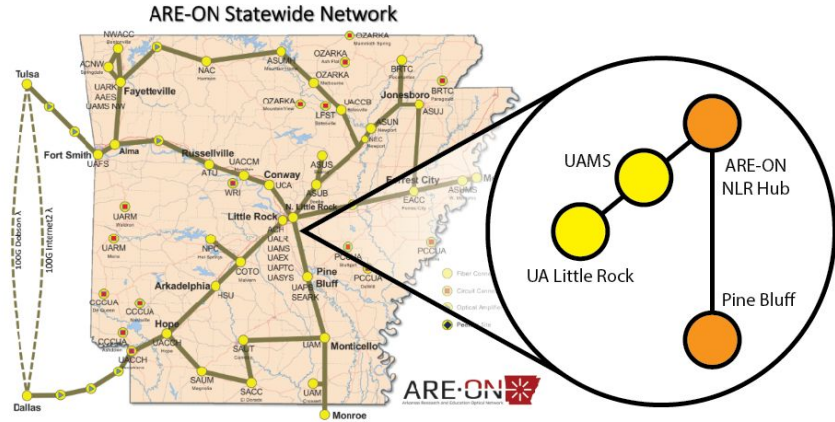
The transition has already begun — what you encrypt today must outlast the arrival of cryptographically relevant quantum computers.

Quantum Security on a Quantum Network

Experimental Quantum Network Between UALR and UAMS

2nd Phase to extend to UAPB

<https://www.gunnect.inc/carina>



Future Work

Quantum Security on the Quantum Network

Multiple Joint Works to Support Entangled Bit Transmission?

How do the concepts of **P**ost-**Q**uantum **C**ryptography
and **Q**uantum **K**ey **D**istribution
be coexisted into the emerging
Quantum Network of entangled qubits?

What security primitives are required to protect routing, authentication, and entanglement management in future quantum Internet infrastructures?

Justification for Future Work

Quantum Security on the Quantum Network

Key methods and use cases for quantum security on these networks include:

- Post-Quantum Cryptography (PQC)
- Quantum Key Distribution (QKD)
- Layered Security Protocols
- Compliance and National Mandates
 - Ensure "quantum readiness" and "cryptographic agility" the ability of a state's systems to rapidly adopt new security protocols as quantum computers begin to challenge existing encryption
- Educational Foundation

Quantum Security on the Quantum Network

Post-Quantum Cryptography (PQC)

- PQC is essential for preparing an organization's security framework for "crypto-agility".
- It involves developing algorithms and readiness assessments to ensure that even as quantum computing power grows, encrypted data remains safe from decryption by future quantum adversaries

Quantum Security on the Quantum Network

Quantum Key Distribution (QKD)

- This is a primary security solution that enables the distribution of encryption keys with theoretical absolute security
- Practical implementations have already demonstrated QKD over live commercial metro fiber networks, bridging the gap between research and real-world infrastructure

Quantum Security on the Quantum Network

Layered Security Protocols

- In a "quantum network stack," security is integrated at multiple levels to ensure robust protection
 - **Link Layer:** Uses hardware authentication to verify the identity of nodes on the network
 - **Application Layer:** Employs authentication-aware protocols to manage the secure transfer of information gained from quantum state measurements

Quantum Security on the Quantum Network

Compliance and National Mandates

- Implementing these encryption methods ensures that organizations meet evolving national quantum cryptography mandates
- This is particularly useful for enterprise customers who must scale their operations safely across "beyond boundaries" networks

Quantum Security on the Quantum Network

Educational Foundation

- Teaching these concepts, such as superposition, uncertainty, and QKD, is a core goal of the emerging **Arkansas Quantum Curricula**, ensuring the future workforce understands how to implement and maintain these secure networking protocols
- By integrating these specialized security solutions, a quantum network can move beyond local simulations to support **real-world applications** in highly sensitive fields like healthcare and government research

Suggestions for new direction research for Quantum Security

- **Post-quantum digital signatures and key-establishment mechanisms** : authenticate quantum nodes, repeaters, controllers, and routing messages on the classical control plane.
- **QKD or quantum-derived key material**: information-theoretic session keys where quantum channels are available.
- **Secure routing and access-control primitives** : prevent malicious route manipulation, impersonation, and unauthorized use of entanglement resources.
- **Entanglement-verification and fidelity-checking mechanisms** : detect tampering, decoherence, or compromised repeaters during entanglement generation, swapping, and distribution.
- **auditability, metadata privacy, and threshold/trust-management mechanisms** : quantum networks will still rely on classical orchestration and partially trusted intermediate nodes.